Incident investigation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Incident investigation before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Monitoring quality drift

Without opening the old lesson, explain one key idea from Monitoring quality drift. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Deployment environments

Without opening the old lesson, explain one key idea from Deployment environments. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Data retention and deletion

Without opening the old lesson, explain one key idea from Data retention and deletion. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

> Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

Incident investigation preserves a timeline and distinguishes **observation** from hypothesis. Repair before **evidence** collection can erase clues.

Record timestamps, versions, requests, authorised actions and outcomes with minimised logs. Separate immediate **containment** from diagnosis. Compare candidate explanations against **evidence**.

Practise with fading help

> Complete the supported Incident investigation trace and record the first step you can explain without help.

Use W2: Trace the worked case for Incident investigation. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

Individual reasoning check

> Check the proposed Incident investigation explanation against the supplied facts before discussing it.

Use the worked case for Incident investigation. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

10:00 permissions config changes; 10:03 restricted read; model version unchanged. Name a plausible cause and one test.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Incident investigation facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Trace the worked case

> Trace the worked case. Apply the supplied Incident investigation facts and keep your reasoning.

At09:00 corpus updates; 09:05 stale cache response; 09:06 cache disabled. Team says model weights changed, but weight version is unchanged. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Audit the limit

> Audit the limit. Apply the supplied Incident investigation facts and keep your reasoning.

A report omits this qualification: A plausible narrative needs controlled or record-based confirmation and must preserve unknowns.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Incident investigation facts and keep your reasoning.

10:00 permissions config changes; 10:03 restricted read; model version unchanged. Name a plausible cause and one test.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Incident investigation facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Incident investigation to Grounded Learning Service, identify a limit and choose the next useful test.

Add a labelled incident investigation **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of incident investigation.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**
>
> _____
>
> _____
>
> _____
>
> _____

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Incident investigation

> An incident timeline. After a schedule update, a service gives an old classroom location. Explain the labelled step before checking the changed condition.

An **incident timeline**. After a schedule update, a service gives an old classroom location.

14:00 version 6 publishes room D; 14:05 answer uses version 5 room C; 14:06 user reports mismatch.

AI ACADEMY / AI-6/26

Incident investigation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Order events

The new source predates the request and report.



2. Locate stale evidence

The answer trace identifies version 5 rather than current version 6.



3. Test explanation

Reproduce version selection with controlled fixtures before naming the exact bug.

CHANGE ONE THING

Trace instead shows version 6 retrieved and room C drafted.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Order the three incident records and locate the old **evidence** version. How does the investigation change if the trace instead retrieves version 6 but drafts room C?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Grounded Learning Service record.

Project: Grounded Learning Service

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

An **incident timeline**. After a schedule update, a service gives an old classroom location.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Reproduce version selection with controlled fixtures before naming the exact bug.

1. Test explanation

2. Order events

3. Locate stale **evidence**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Test explanation Reproduce version selection with controlled fixtures before naming the exact bug. A timeline separates observed events from a proposed cause.

Evidence can locate a mismatch without yet explaining why it happened.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Trace instead shows version 6 retrieved and room C drafted. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. Investigate drafting or validation before blaming caching.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Investigate drafting or validation before blaming caching. The mismatch now appears after retrieval.

Repair a missing building block

Review Monitoring quality drift (ai-6/25). Goal: Explain the **mechanism** and terminology of monitoring quality drift. Short practice: Compute both clear and ambiguous slice rates for A and B.

Which slice weakens, and how should that slice be reported when no ambiguous questions are observed?. Return to this week and explain what you changed.

Review Deployment environments (ai-6/23). Goal: Explain the **mechanism** and terminology of deployment environments. Short practice: Identify the staging data and credential boundary.

Why must the rehearsal stop if its credential reaches production even though the **process** is labelled staging?. Return to this week and explain what you changed.

Review Data retention and deletion (ai-6/20). Goal: Explain the **mechanism** and terminology of data retention and deletion. Short practice: Apply the seven-day boundary to T1 and T2.

How many T2 locations need verification, and may deletion be called complete if its backup remains?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

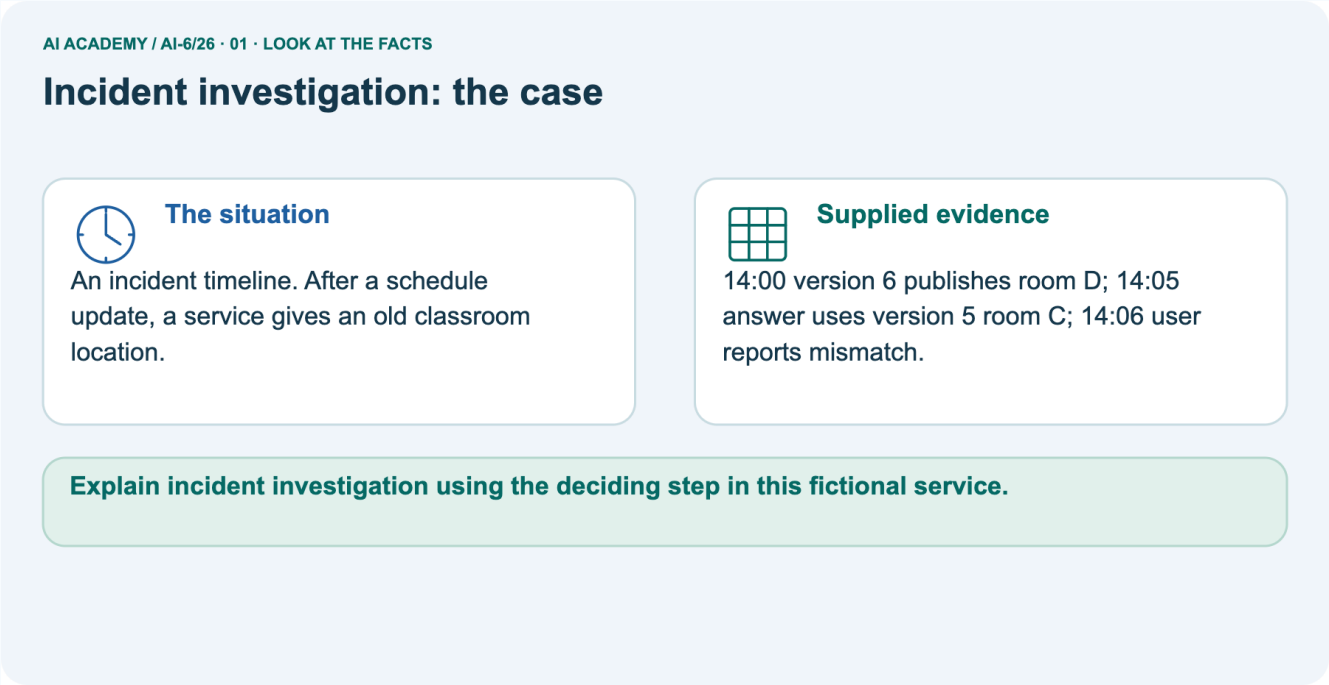
> **What I repaired, the evidence I kept and the next test**
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> |---|
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> | |
> | |

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Incident investigation picture from its labels, then record a relevant limit.

01. Incident investigation: the case



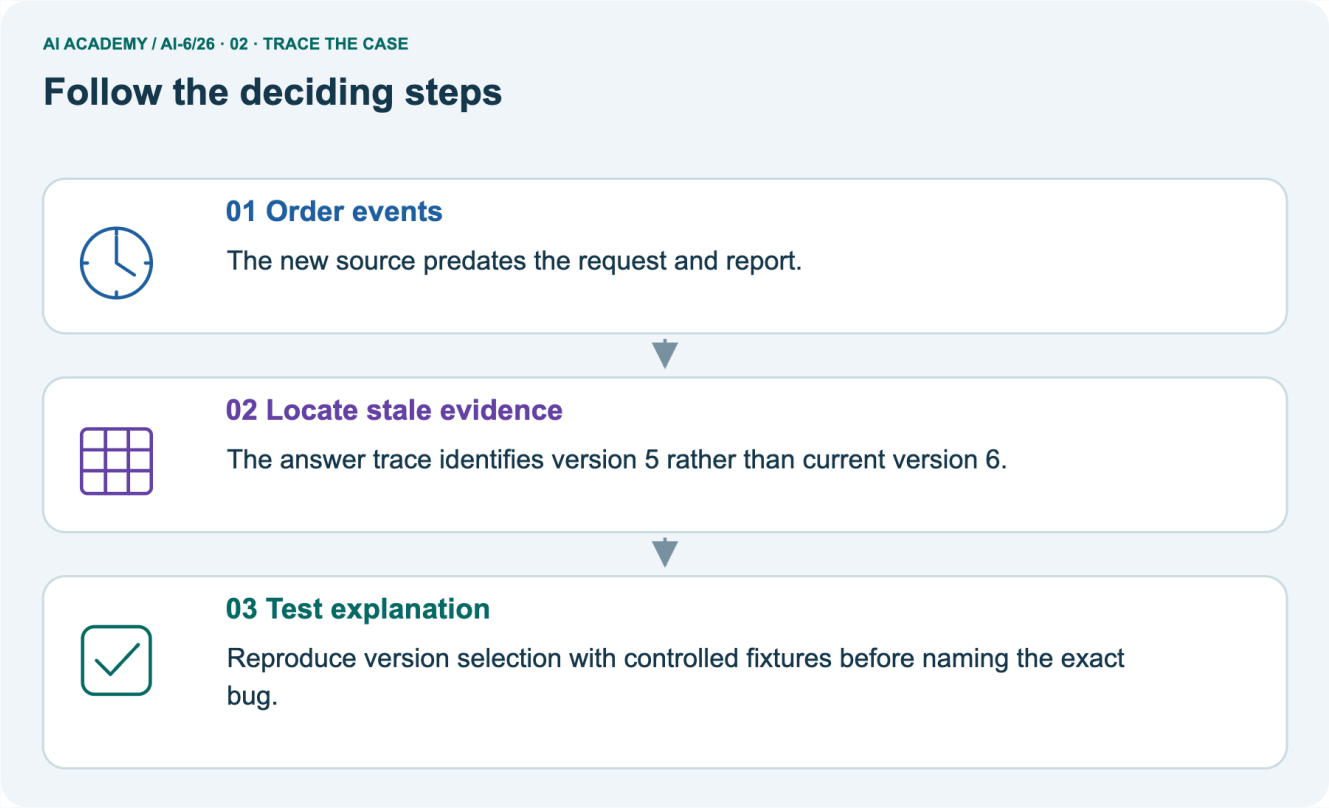
Follow the labelled picture with the matching explanation. Open the vector diagram

An **incident timeline**. After a schedule update, a service gives an old classroom location. Supplied facts: 14:00 version 6 publishes room D; 14:05 answer uses version 5 room C; 14:06 user reports mismatch.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



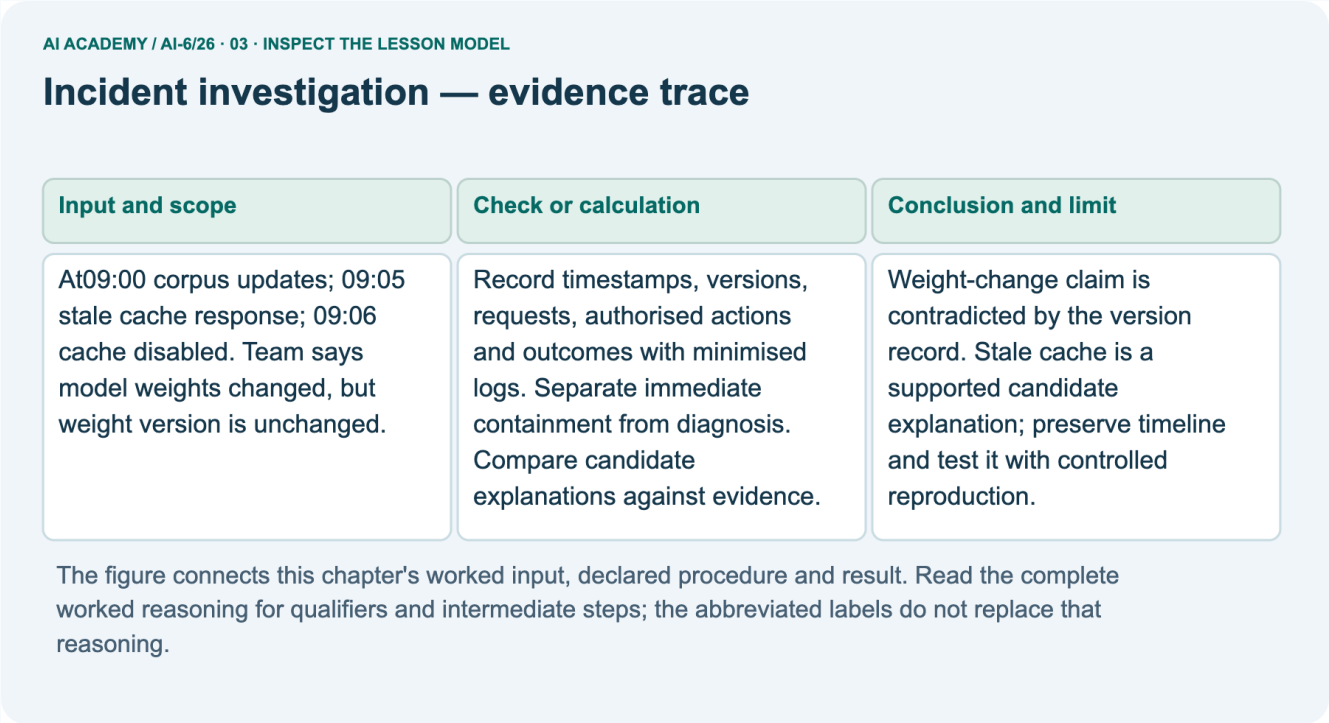
Follow the labelled picture with the matching explanation. Open the vector diagram

Order events: The new source predates the request and report. → Locate stale **evidence**: The answer trace identifies version 5 rather than current version 6. → Test explanation: Reproduce version selection with controlled fixtures before naming the exact bug.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Incident investigation — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Incident investigation — **evidence** trace. At09:00 corpus updates; 09:05 stale cache response; 09:06 cache disabled.

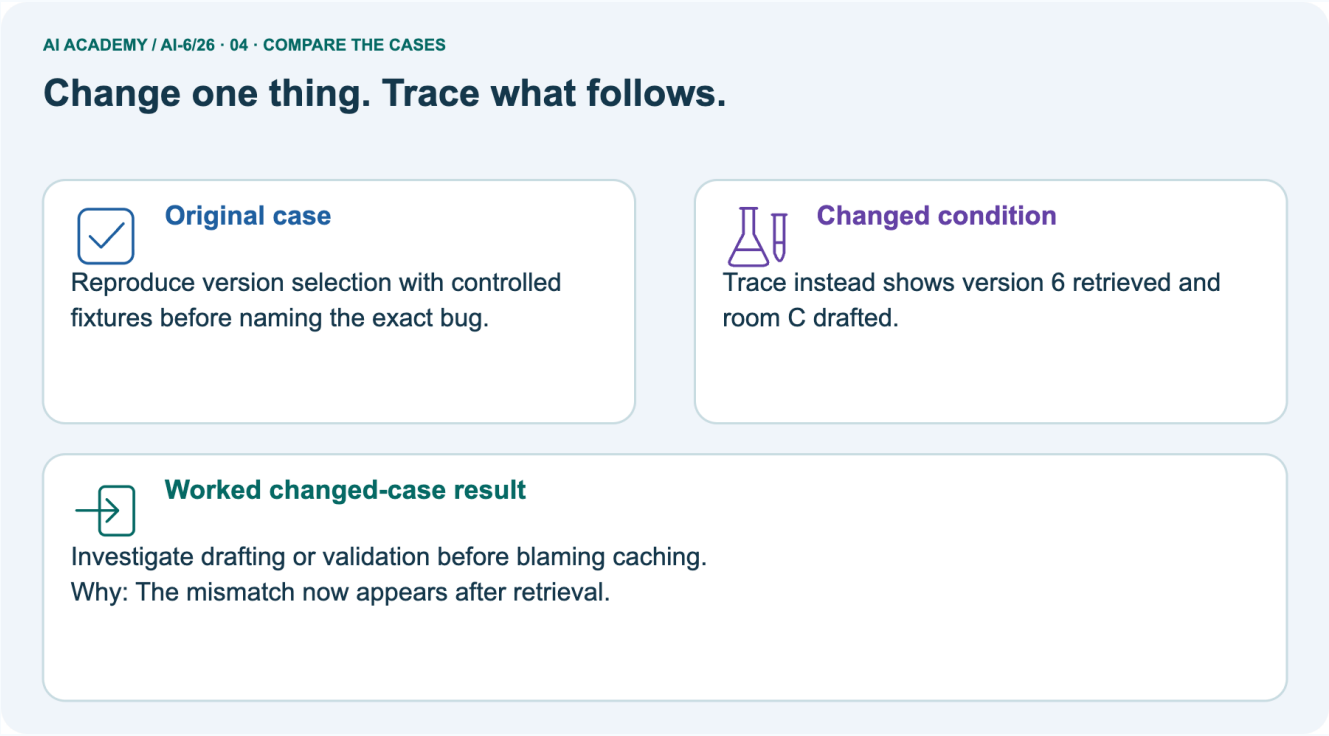
Team says model weights changed, but weight version is unchanged. | Record timestamps, versions, requests, authorised actions and outcomes with minimised logs. Separate immediate **containment** from diagnosis.

Compare candidate explanations against **evidence**. | Weight-change claim is contradicted by the version record. Stale cache is a supported candidate explanation; preserve timeline and test it with controlled reproduction.

> Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

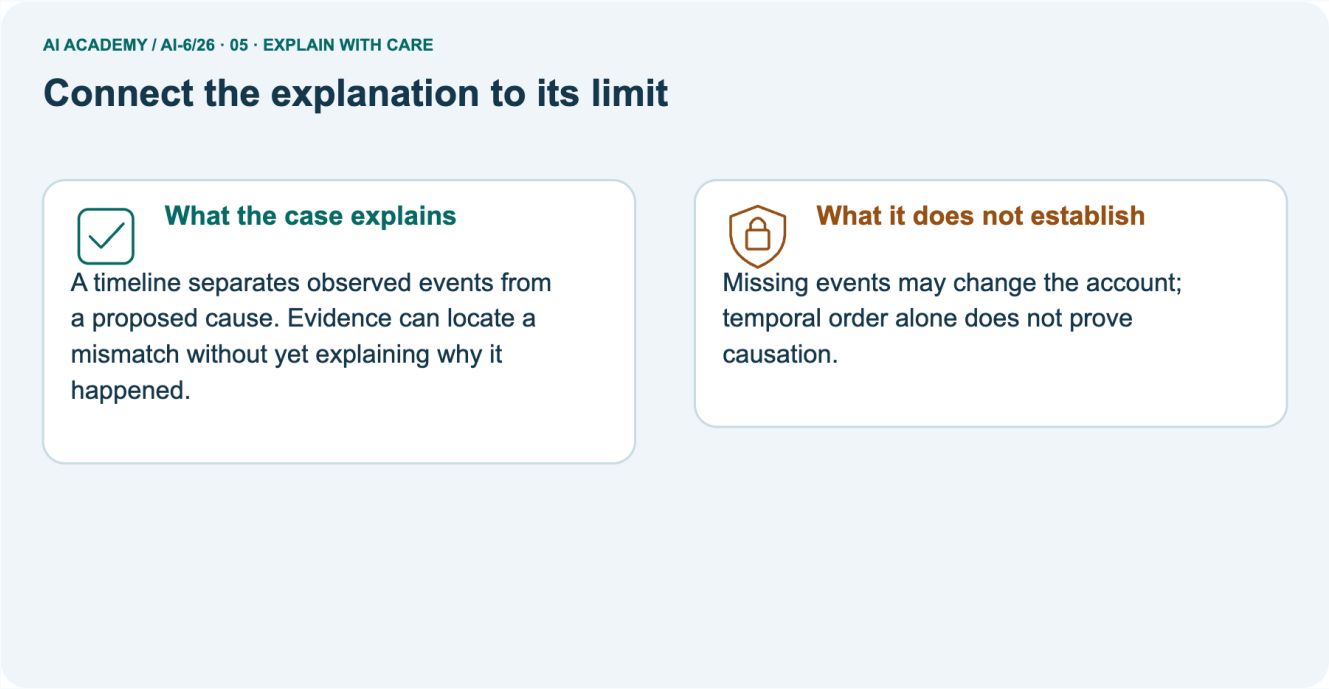
Original case: Reproduce version selection with controlled fixtures before naming the exact bug.
Changed condition: Trace instead shows version 6 retrieved and room C drafted.

Expected result: Investigate drafting or validation before blaming caching. Reason: The mismatch now appears after retrieval.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



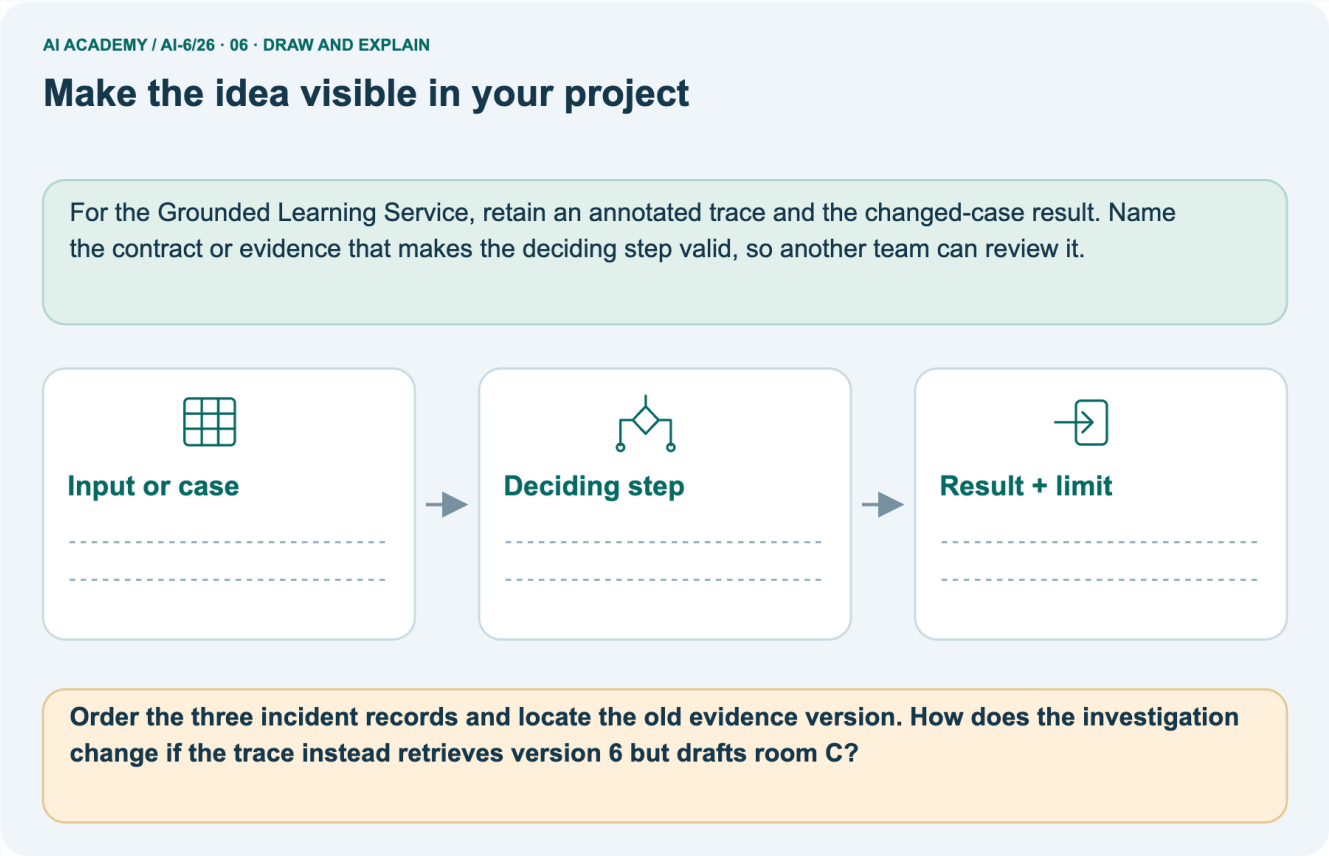
Follow the labelled picture with the matching explanation. Open the vector diagram

A timeline separates observed events from a proposed cause. **Evidence** can locate a mismatch without yet explaining why it happened. Limit: Missing events may change the account; temporal order alone does not prove causation.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Order the three incident records and locate the old **evidence** version.

How does the investigation change if the trace instead retrieves version 6 but drafts room C?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.